工作坊 2：教師準備及流程

Workshop 2: preparation and flow

每堂 60 分鐘；兩人一組。目標是讀散點圖、執行程式及解釋預測。
60 minutes, pairs. Aim to read scatter plots, run code and explain a prediction.

課前：確認學校 Google 帳戶可用 Colab；檢查網絡及鍵盤裝置。
Before class: check approved Google accounts, Colab access, internet and keyboards.

Notebook 有兩個獨立程式格，每節只需執行一格；工作紙印有完整程式。
The notebook has two independent code cells. The worksheet prints all the code.

程式已在本機執行檢查；仍需老師在實際課室裝置試開 Colab。
Code was executed locally; rehearse Colab access on classroom devices.

0–10 分鐘：用第一堂的散點回顧趨勢；示範儲存副本及播放按鈕。
0–10: recall trends from Workshop 1; show Save a copy and the play button.

10–25 分鐘：執行第 1 格，看外送距離與收費，回答兩條觀察問題。
10–25: run cell 1, explore delivery distance and fee, and answer two questions.

25–35 分鐘：換成時間及成績數據，口頭重溫 m 及 c，再介紹 fit。
25–35: switch to study time and scores, recall m/c orally, then introduce fit.

35–50 分鐘：執行第 2 格，找模型摘要、直線及紫色預測點，記下分數。
35–50: run cell 2; locate the summary, line and purple prediction; record the score.

50–60 分鐘：完成直線的觀察問題，分享預測是估計而非保證。
50–60: complete the line observation and discuss estimates versus guarantees.

參考：外送收費整體上升，但 2→3、4→5、6→7 公里都有下降。
Reference: fees generally rise; dips occur from 2→3, 4→5 and 6→7 km.

模型摘要：7 筆觀察；y = 2x + 50。每天 4.5 小時的預測爲 59 分。
Model summary: 7 observations; y = 2x + 50. Prediction at 4.5 h/day: 59 marks.

直線不通過每一點。接受合理描述；不要求手算或掌握每行程式。
The line does not pass through every dot. Accept descriptions; no calculation required.

以 # 開頭的行是註解。改數據後，重新執行該節整格；每格包含所需工具。
Lines starting with # are comments. After edits, rerun the whole cell with its tools.

成績爲模擬；預測不是保證，也不能用此數據證明因果。
Scores are simulated. Predictions are not guarantees or evidence of cause.

若學生不能登入，可共用獲准的裝置，或用老師示範及工作紙。
If access fails, use an approved shared device or teacher demo with the worksheet.

Resources: gordany.github.io/f2-linear-regression/workshop-2/

工作坊 3：教師準備及流程
Workshop 3: preparation and flow

課前：準備筆、擦膠、空背景；先試相機權限及 Teachable Machine。
Before class: prepare pens, erasers and empty backgrounds; test camera access and the site.

由老師預先拍攝 24 張測試相片，分 A／B，均不放入訓練圖片。
Teacher prepares 24 test photos in A/B folders, kept separate from all training images.

每批：筆 4 張、擦膠 4 張、只有背景 4 張；B 用新的拍攝，不是複製 A。
Each set: 4 pen, 4 eraser, 4 background-only photos. B uses fresh shots, not copies of A.

A／B 採相近條件：正面、斜角、較遠、新背景；背景類須沒有物件。
Use similar conditions in A/B: front, angled, farther away and another background.

檔案名例如 A/pen_01.jpg、B/pen_01.jpg；另有 eraser_01、empty_01 至 04。
Name files by set/class, e.g. A/pen_01.jpg and B/pen_01.jpg; number each class 01–04.

派 A／B 資料夾給每組。B 保持未用，改善後才以 Preview 的圖片輸入測試。
Distribute A/B folders. Keep B unused until final testing via Preview's image input.

0–8 分鐘：藏物件／換背景小遊戲，口頭分享看到的線索。
0–8: hidden-object/background game and oral discussion of visual clues.

8–20 分鐘：圖像、CNN 的想法，建立三類，解釋預留 A／B。
8–20: image/CNN ideas, set up three classes and explain A/B.

20–35 分鐘：各類目標 20 張以上有變化的訓練圖片，再訓練。
20–35: aim for 20+ varied training images per class, then train.

35–55 分鐘：測 A、補充例子、重訓、測未用 B；每類各記 4 張。
35–55: test A, add examples, retrain, test fresh B; record counts out of4.

55–60 分鐘：反思模型分數、新圖片測試及未教物件。
55–60: reflect on model scores, new-image testing and unfamiliar objects.

A 全部答對可寫「未見錯誤」。不要造錯誤，也不要提早用 B。
If A is all correct, write "none found". Do not invent errors or use B early.

20 張是課堂目標，不是產品門檻。模型分數不等於測試準確率。
20 images is a classroom target, not a product minimum. Scores are not test accuracy.

借用預訓練圖像特徵，建立自己的分類部分；不需從零訓練整個 CNN。
Reuse pretrained image features and teach the classifier; no full CNN training from scratch.

只拍物件。若相機不可用，用老師備妥的訓練相片；A／B 仍須分開。
Use object photos. If camera access fails, supply training photos while keeping A/B separate.

Resources: gordany.github.io/f2-linear-regression/workshop-3/